

Green Corridor for Ambulances

A working simulation of Kuwait City's traffic signals clearing the road ahead of an emergency vehicle — automatically, safely, and back to normal the moment it has passed.

Built on real OpenStreetMap roads of Kuwait City · Eclipse SUMO traffic microsimulation · Presented to the Ministry, August 2026

The problem: red lights cost ambulances minutes

Today an ambulance with its lights on still meets the same signals as everyone else.

1

Waiting at the red

On a typical downtown run an ambulance meets 8 signals. Each red adds tens of seconds while the patient waits.

2

Nowhere to go

Even when drivers want to give way, a full queue at a red light has no space to move aside.

3

Crossing on red is unsafe

Driving through a red light against live traffic is the most dangerous moment of the whole journey.

Our goal: the signal ahead is already green when the ambulance reaches it — and the traffic in front has already moved.

What we built: a living model of Kuwait City's roads

Not a drawing — a simulation where every car, signal and ambulance behaves second by second.

1

Real roads

Downtown Kuwait City is taken directly from OpenStreetMap: one-way streets, turn rules and 78 signalized junctions. A second model covers all six governorates (223 junctions).

2

Real traffic behaviour

Built on Eclipse SUMO, the traffic simulator used by transport authorities worldwide. Demand follows the Kuwaiti weekday pattern, hour by hour — 07:00 peak, near-empty streets at 03:00.

3

Built for people, not engineers

A live map for the control room, a phone screen for the ambulance driver, an operations log for supervisors, and an animated explainer in Arabic and English.

78

signalized junctions, downtown model

223

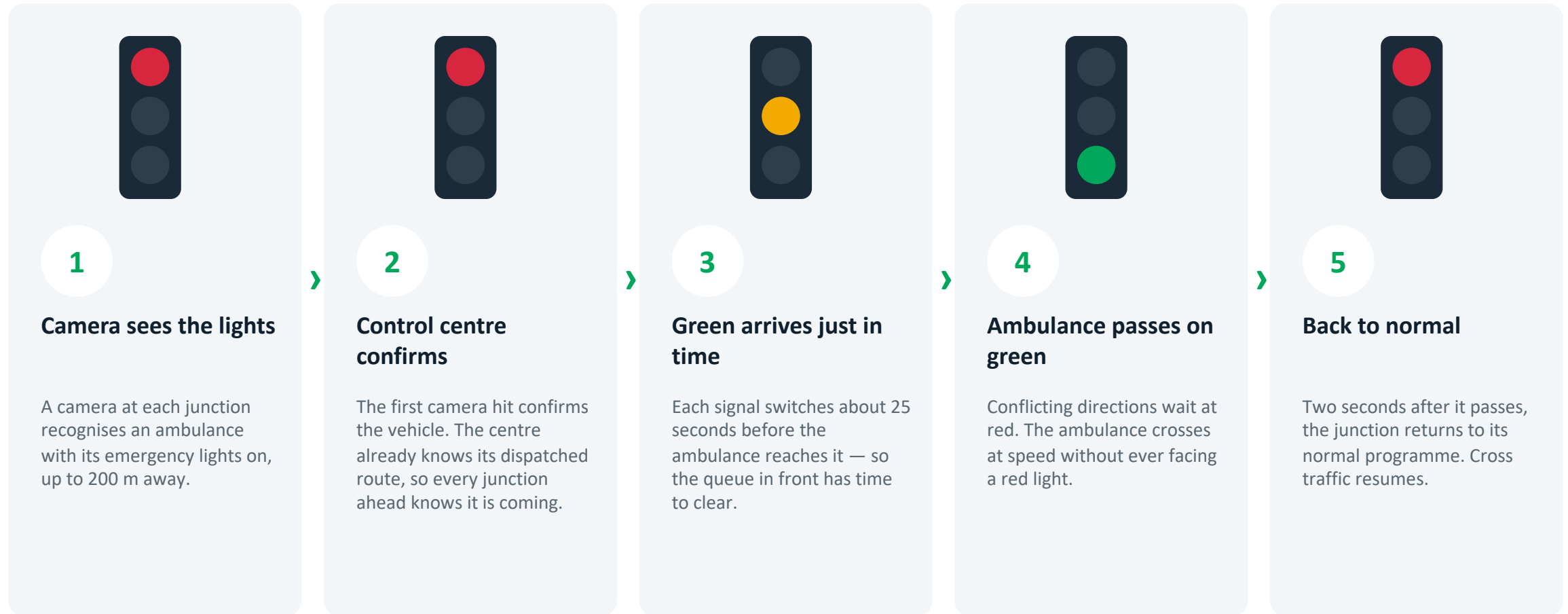
junctions, six-governorate model

1 hospital

per governorate, real MoH sites

How it works: five steps, then back to normal

Every step happens automatically. No one in the ambulance or control room presses a button.



Safety first: what the system will never do

A green corridor is only acceptable if it is at least as safe as a normal signal. These rules are built in.



Never a sudden switch

Conflicting traffic always gets 3 seconds of amber, then 2 seconds of all-red, before the corridor turns green. Cars already inside the junction have time to leave.



Never a dark signal

Signals are never switched off. Every junction always shows a proper red, amber or green to every direction.



Never held too long

A single hold is capped at 90 seconds. After that, cross traffic is guaranteed a normal cycle — unless the ambulance is already at the stop line.



Never boxed in

While the ambulance is physically inside the junction, the green is kept. A signal can never return to red with the ambulance halfway across.

Beyond the green light: the intelligence around it

Six capabilities that turn a signal trick into an operating system for emergencies.

1

Own route planner

The fastest route is computed on live travel times, respecting one-way streets and turn rules. The driver's screen and the signals follow the same route.

2

Scene, then hospital

On arrival the corridor pauses while the patient is loaded, then a new route opens automatically to the nearest hospital by travel time.

3

Learns the traffic

Every road is modelled as a small probability model that learns continuously which streets jam and when — and routes around the predicted jam.

4

Two ambulances, one junction

A published decision matrix rules which ambulance goes first. Unclear cases are referred to a human supervisor with one-click buttons.

5

Speed exemption, audited

With lights on, the ambulance may exceed the limit; every speed camera logs the pass as 'NO CITATION' so nothing is left to dispute.

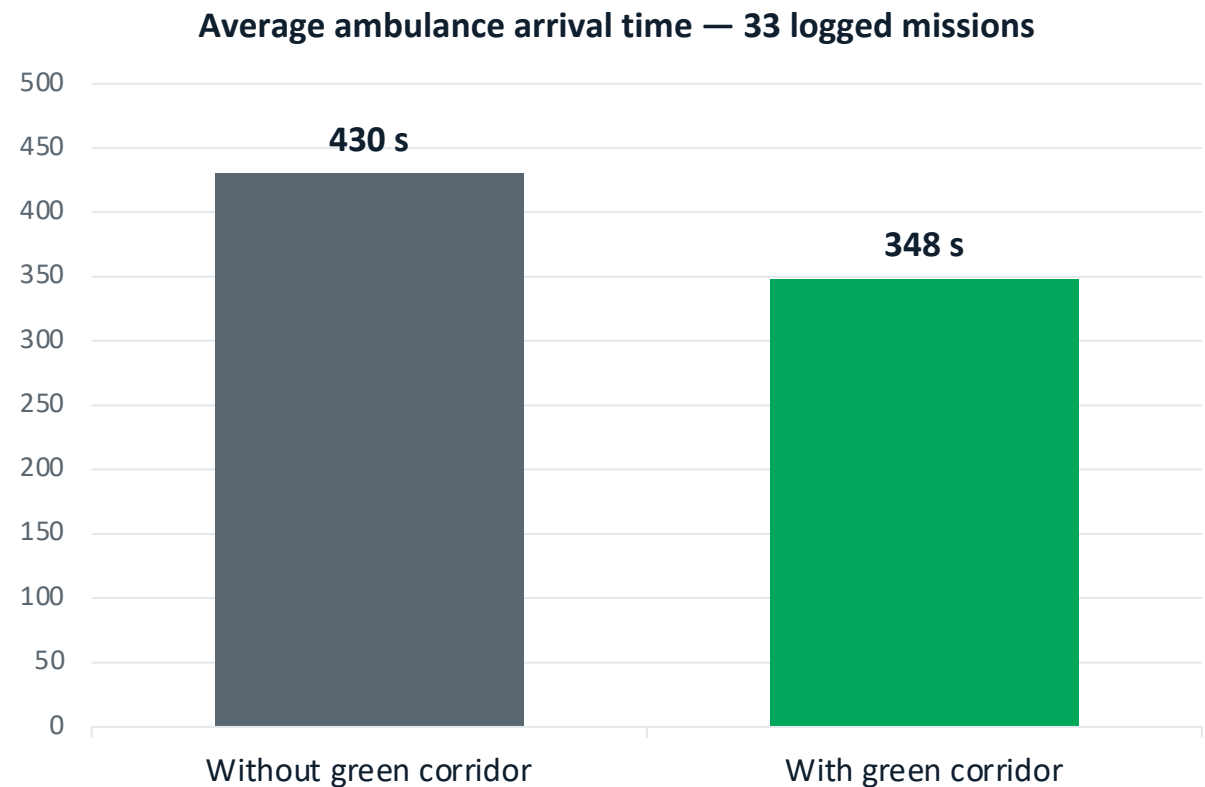
6

Ask the record

Supervisors ask questions in plain language — 'what happened on case A-012?' — and get an answer with citations from the operations log.

What we measured

Same streets, same traffic, same dispatches — with the green corridor and without it.



19% faster

mission log, 33 runs

About 80 seconds saved on an average run; the ambulance never stops at a signal.

+9.1% speed

measured at peak demand

Mean speed of all traffic rose when junctions also give early green to a lone waiting approach.

–42.5% halted

measured at peak demand

Fewer vehicles stopped network-wide — other road users gain too, not only the ambulance.

'Without' times are the system's own per-junction estimate of the signal waits the corridor removed, in the same traffic.

Where we go from here

The model is ready. Three things from the Ministry would turn it into a pilot.

1

Real signal timings

The road layout is real; the signal plans are still synthetic. With the actual timing plans, results become a forecast for specific junctions.

2

Real traffic counts

A small file of hourly counts (hour, multiplier) plugs straight into the model and replaces the calibrated profile.

3

One pilot corridor

Pick a single hospital-to-area corridor. We simulate it first, then it can be tried on the street with the existing camera and controller infrastructure.

Everything shown today runs on an ordinary computer, with no internet connection, and logs every decision for after-action review.

Thank you. Questions welcome.